FRONTEND LAYER (React + Vite)

React App (Vite Dev Server)
Port 5173
Hot Module Replacement

Xterm.js Terminal Component
- WebSocket Client
- Keystroke Capture
- ANSI Rendering

React-Three-Fiber Canvas
- Three.js Scene
- 3D Server Model (Cube)
- Color State (Green/Red)
- Animation Loop

AI Chat Interface
- Streaming Response Display
- Markdown Renderer
- Code Syntax Highlighting

WEBSOCKET COMMUNICATION (Gorilla)

WebSocket Connection Manager
- Gorilla WebSocket Upgrader
- Connection Pool (map[sessionID]*conn)
- Ping/Pong Heartbeat (30s)
- Message Queue (buffered chan)

Message Router (Go)
- JSON Message Parser
- Command Type: [EXEC, SNAPSHOT, RESTORE]
- Session ID Validation
- Error Handler

Response Streamer
- Stdout/Stderr Pipe Reader
- Real-time Event Emitter
- Token-by-token AI Stream
- gzip Compression

BACKEND CORE (Golang)

Session Manager
- CreateSession(userID) → sessionID
- GetSession(sessionID) → *Session
- DeleteSession(sessionID)
- Session Timeout (5 min idle)
- Concurrent Sessions Limit (100)

Container Orchestrator
- SpawnContainer(config) → containerID
- Docker SDK / runc Wrapper
- Resource Limits: CPU=0.5, RAM=256MB
- Network Isolation (bridge)
- Volume Mount: /sandbox
- Auto-cleanup on Exit

Command Executor (Airlock Core)
- exec.Command() with Context
- Stdin/Stdout/Stderr Pipes
- Timeout: 30 seconds
- Exit Code Capture
- Signal Handling (SIGKILL, SIGTERM)

OBSERVABILITY & MONITORING (eBPF + Logs)

eBPF Syscall Tracer
- Cilium/eBPF Library
- Tracepoints: sys_enter_kill, sys_exit_group
- Filter: PID=1 or ExitCode!=0
- Perf Event Buffer (1MB)
- Parse Kernel Structs

Log Processor & Deduplicator
- NormalizeLog() → Hash Signature
- Dedup Window: 5 seconds
- Priority Filter (LogCritical, LogError)
- ExtractDiagnostics() → Structured JSON
- Circular Buffer (last 20 events)

CloudWatch Logs Client
- AWS SDK for Go v2
- PutLogEvents() Batch API
- Log Group: /Entropy/sessions
- Retention: 7 days
- Async Write (goroutine)

AI INTELLIGENCE (Amazon Bedrock)

Bedrock Client (Go)
- bedrockruntime.InvokeModelWithResponseStream()
- Model: anthropic.claude-3-5-sonnet-20241022-v2:0
- Region: us-east-1
- Credentials: IAM Role / Access Keys

Prompt Engineering Pipeline
- System Prompt (Entropy Persona)
- Input: {command, exit_code, error_msg, syscall}
- Context Window: 8K tokens
- Temperature: 0.7
- Max Tokens: 500

AI Response Parser
- Stream Token Handler
- Markdown Formatter
- Code Block Extraction
- Error Retry Logic (3 attempts)

PERSISTENCE & STATE (Snapshots + Metadata)

Snapshot Manager
- docker commit → Image
- docker save → .tar.gz
- Compression: gzip Level 6
- Upload to S3: Entropy-snapshots/
- Lifecycle: Delete after 30 days

Metadata Database (PostgreSQL)
- Table: user_sessions
- session_id (UUID PK)
- user_id (VARCHAR)
- command_history (JSONB)
- file_tree (JSONB)
- crash_logs (JSONB)
- created_at, updated_at

Restore Engine
- FetchMetadata(sessionID)
- SpawnFreshContainer()
- InjectFiles() via docker cp
- RestoreHistory() → Replay Commands
- Restore Time: 2-3 seconds

AWS INFRASTRUCTURE (EKS + Services)

Amazon EKS Cluster
- Kubernetes Version: 1.31
- Node Group: t3.medium (2 vCPU, 4GB)
- Auto Scaling: 2-10 nodes
- Namespace: Entropy-sandbox

AWS Amplify Hosting
- React Build Output
- CloudFront CDN
- Auto Deploy from GitHub
- Branch: main

S3 Bucket
- Bucket: Entropy-snapshots
- Versioning: Enabled
- Lifecycle: 30 days expiry
- Encryption: AES-256

RDS PostgreSQL
- Instance: db.t3.micro
- Storage: 20 GB gp3
- Multi-AZ: No (hackathon)
- Backup: 7 days